



Written Examination
Group and Ring Theory
MATP33
Saturday, 29 November 2025
Duration: 8:00–13:00

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Answers

1. The same does not hold for the property of being injective.
3. We have

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (x+3)(x+1)^2 \end{pmatrix}.$$

The rank is 3.